UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 00498

LH QUANTUM MECHANICS 3

SUMMER EXAMINATIONS 2024

Time allowed: 1 hour 30 minutes

*Answer **two** questions. If you answer more than two questions, credit will only be given for the* best *two answers.*

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

A formula sheet that may be required will be found at the end of this question paper.

No calculator is permitted in this examination.

*Answer any **two** questions. If you answer more than **two** questions, credit will only be given for the **best** two answers.*

1. (a) Prove that the time evolution of the expectation value $\langle \hat{A} \rangle$ of a time-independent operator $\hat{A}$ in a system with Hamiltonian $\hat{H}$ is described by the Ehrenfest equation,

$$\frac{\mathrm{d}\langle \hat{A} \rangle}{\mathrm{d}t} = \frac{i}{\hbar} \left\langle [\hat{H}, \hat{A}] \right\rangle .$$ **[4]**

(b) Explain what is meant by a stationary state and write down (or derive) its time-dependence. Show that $\langle \hat{A} \rangle$ is time-independent in any stationary state. **[6]**

An operator $\hat{A}$, which does not commute with $\hat{H}$, has eigenstates $|a_1\rangle$, $|a_0\rangle$ and $|a_{-1}\rangle$ corresponding to eigenvalues $a_{\pm 1} = \pm 1$ and $a_0 = 0$. The eigenstates $|n\rangle$ of $\hat{H}$, where $n = \pm 1,\ 0$, have the energies $\pm\hbar\omega,\ 0$, respectively. The states $|a_n\rangle$ are expressed via $|n\rangle$ as follows:

$$|a_{\pm 1}\rangle = \tfrac{1}{2}\Big(|1\rangle \pm \sqrt{2}\,|0\rangle + |-1\rangle \Big)$$

$$|a_0\rangle = \tfrac{1}{\sqrt{2}}\Big(|1\rangle - |-1\rangle \Big) .$$

At time $t = 0$ a measurement of $\hat{A}$ is made and it yields a value $+1$.

(c) Write down the system state $|\psi\rangle$ at a time $t \geqslant 0$. **[5]**

(d) Hence show that the probability that a measurement of $\hat{A}$ would yield the value $a_1 = 1$ at $t \geqslant 0$ is given by

$$P_1(t) = \tfrac{1}{4}\big(1 + \cos\omega t\big)^2 .$$ **[5]**

(e) Find similar expressions for the probabilities $P_{-1}(t)$ and $P_0(t)$ that a measurement of $\hat{A}$ would yield its eigenvalues -1 or 0, respectively. **[6]**

(f) Use the results of parts (d) and (e) to calculate $\langle \hat{A} \rangle$ as a function of t. **[4]**

2. (a) Show that matrix representations of spin operators $\hat{S}_{x,y}$ for spin 1 in the basis where $\mathsf{S}_z = \hbar\,\mathrm{diag}\,(1,\,0,\,-1)$ are given by

$$\mathsf{S}_x = \frac{\hbar}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \text{and} \quad \mathsf{S}_y = \frac{i\hbar}{\sqrt{2}}\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$ **[9]**

[*Hint: Use matrix elements of ladder operators $\hat{L}_\pm$ and the correspondence of orbital angular momentum operators to spin operators given in Formula Sheet.*]

The energy levels of an ion with spin $s = 1$ can be described in some crystals with the following spin Hamiltonian:

$$\hat{H} = \alpha\hat{S}_z^2 + \beta\left(\hat{S}_x^2 - \hat{S}_y^2\right) \text{ where } \alpha,\,\beta > 0\,, \quad \alpha \neq \beta\,.$$

(b) Show that $\hat{H}$ is represented in the same basis as in part (a) by the matrix

$$\mathsf{H} = \hbar^2\begin{pmatrix} \alpha & 0 & \beta \\ 0 & 0 & 0 \\ \beta & 0 & \alpha \end{pmatrix}.$$ **[6]**

(c) Use the above matrix representation to solve the eigenvalue problem for $\hat{H}$, i.e. to find its eigenenergies and eigenstates. **[9]**

(d) For a given α, sketch an energy-level diagram as a function of β. What is the groundstate energy for $\beta = \frac{1}{2}\alpha$ and for $\beta = 2\alpha$? **[6]**

3. A Hamiltonian of a particle of mass m in a parabolic potential well (the quantum harmonic oscillator) can be represented as

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2} \equiv \hbar\omega\left(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\right) ,$$

where the mutually adjoint ladder operators $\hat{a}$ and $\hat{a}^\dagger$ are expressed via the position and momentum operators $\hat{x}$ and $\hat{p}$ as given in Formula Sheet.

(a) Calculate the expectation values $\langle n|\hat{x}\,|n\rangle$, $\langle n|\hat{p}\,|n\rangle$ and $\langle n|\hat{x}^2\,|n\rangle$, $\langle n|\hat{p}^2\,|n\rangle$ and show that they are consistent with the Heisenberg uncertainty principle $\Delta x\Delta p \geqslant \hbar/2$. **[8]**

(b) Show that the normalised groundstate wavefunction is given by

$$\psi_0(x) \equiv \langle x\,|\,0\rangle = (\pi\ell^2)^{-\frac{1}{4}}\,\mathrm{e}^{-\frac{x^2}{2\ell^2}}, \quad \text{where } \ell^2 \equiv \hbar/(m\omega)\,. \qquad \textbf{[6]}$$

(c) Find the wavefunction of the 1st excited state, $\psi_1(x) \equiv \langle x\,|\,1\rangle$. **[4]**

[*Hint: You may find it helpful to use dimensionless operators $\hat{P} \equiv \ell\hat{p}/\hbar$ and $\hat{X} \equiv \hat{x}/\ell$.*]

Next, the particle is placed in a modified potential well with the Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) \text{ with } V(\hat{x}) = \begin{cases} \frac{1}{2}\hbar\omega\frac{|\hat{x}|}{L}, & |x| \leqslant \frac{\ell^2}{L} \\ \frac{1}{2}m\omega^2\hat{x}^2, & |x| \geqslant \frac{\ell^2}{L} \end{cases}.$$

(d) Sketch the potential $V(x)$ and identify the conditions under which the difference $\hat{H} - \hat{H}_0$ to may be treated as a small perturbation. **[4]**

(e) Use the results of part (b) to show that the first-order perturbative correction to the groundstate energy, ε_0, of the unperturbed Hamiltonian $\hat{H}_0$ has the following relative size:

$$\frac{\varepsilon_0^{(1)}}{\varepsilon_0} = \frac{1}{3\sqrt{\pi}}\left(\frac{\ell}{L}\right)^3 . \qquad \textbf{[8]}$$

[*Hint: Argue that in calculating the perturbative correction you can replace $\psi_0(x)$ by a constant.*]

Formula Sheet

Resolution of unity (completeness)
$\hat{I} = \sum_n |\eta_n\rangle\langle\eta_n| = \int_{-\infty}^{\infty} \mathrm{d}x\, |x\rangle\langle x|$

Orthonormality: $\langle \eta_n \,|\, \eta_m \rangle = \delta_{nm}$

Hermitian conjugation: if $|\varphi\rangle = \hat{A}\,|\psi\rangle$, then $\langle\varphi| = \langle\psi|\hat{A}^\dagger$.

Ket and bra relations to the wavefunction:
$\langle x \,|\, \psi\rangle = \psi(x)\,; \quad \langle \psi \,|\, x\rangle = \psi^*(x)$.

Heisenberg uncertainty principle
$\Delta A \Delta B \geqslant \frac{1}{2}\left|\left\langle [\hat{A}, \hat{B}] \right\rangle\right|$,
$\Delta A \equiv \sqrt{\langle \hat{A}^2\rangle - \langle \hat{A}\rangle^2}$

Time evolution $|\psi(t)\rangle = \hat{U}(t)\,|\psi(0)\rangle$,
$\hat{U}(t) \equiv \mathrm{e}^{-i\hat{H}t/\hbar}$

Evolution of operators in Heisenberg picture
$\hat{A}_{\mathrm{H}}(t) = \hat{U}^\dagger(t)\hat{A}(0)\hat{U}(t)$.

Ladder operators for SHO, $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2}$:

$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{\ell} + \frac{i\ell}{\hbar}\hat{p}\right) \equiv \frac{1}{\sqrt{2}}\left(\hat{X} + i\hat{P}\right)$, $\ell \equiv \sqrt{\frac{\hbar}{m\omega}}$,

$\hat{X} = \frac{1}{\sqrt{2}}\left(\hat{a} + \hat{a}^\dagger\right),\ \hat{P} = \frac{i}{\sqrt{2}}\left(\hat{a}^\dagger - \hat{a}\right)$

Commutation relations for $\hat{a}$, $\hat{a}^\dagger$, and $\hat{N} \equiv \hat{a}^\dagger\hat{a}$:
$[\hat{a}, \hat{a}^\dagger] = 1\,, \quad [\hat{a}, \hat{N}] = \hat{a}\,, \quad [\hat{a}^\dagger, \hat{N}] = -\hat{a}^\dagger$.

Action on eigenstates of $\hat{N}$:
$\hat{a}\,|n\rangle = \sqrt{n}\,|n{-}1\rangle,\ \hat{a}^\dagger\,|n\rangle = \sqrt{n{+}1}\,|n{+}1\rangle$.

Angular momentum commutation relations
$[\hat{L}_x, \hat{L}_y] = i\hbar L_z$, $[\hat{L}_y, \hat{L}_z] = i\hbar L_x$,
$[\hat{L}_z, \hat{L}_x] = i\hbar L_y$; $[\hat{L}_k, \hat{L}^2] = 0,\ k = x, y, z$.

Orbital angular momentum eigenstates
$\hat{L}^2\,|\ell, m\rangle = \hbar^2\ell(\ell+1)\,|\ell, m\rangle$,
$\hat{L}_z\,|\ell, m\rangle = \hbar m\,|\ell, m\rangle$
$\ell = 0, 1, 2, \ldots;\ m = 0, \pm1, \ldots, \pm\ell$.

Ladder operators $\hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y$,
$[\hat{L}_+, \hat{L}_-] = 2\hbar\hat{L}_z$, $[\hat{L}_z, \hat{L}_\pm] = \pm\hbar\hat{L}_\pm$
$\hat{L}_\pm\,|\ell, m\rangle = \hbar c^\pm_{\ell,m}\,|\ell, m\pm1\rangle$
$c^\pm_{\ell,m} = \sqrt{(\ell\mp m)(\ell\pm m+1)}$
where integer $|m| \leqslant \ell$ and $|m\pm1| \leqslant \ell$.

Spin has the same properties when replacing $\hat{L}_i \to \hat{S}_i,\ \ell \to s,\ m \to \sigma$ and allowing s and σ being either integer or half-integer

Spin $\frac{1}{2}$ with $\hat{S}_k = \frac{1}{2}\hbar\hat{\sigma}_k$ where Pauli matrices

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$
$$\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Dirac notations for spin $\frac{1}{2}$:
$\left|\frac{1}{2}, \frac{1}{2}\right\rangle \equiv |+\rangle \equiv |\uparrow\rangle,\ \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \equiv |-\rangle \equiv |\downarrow\rangle$.

Density matrix:
$$\hat{\rho} = \sum_n p_n\,|\psi_n\rangle\,\langle\psi_n|$$
where p_n is the classical probability of a system being in a pure state $|\psi_n\rangle$.

Von Neumann entropy
$S = -\operatorname{Tr}[\hat{\rho}\log_2\hat{\rho}]$.

Time-independent perturbation theory
$(\hat{H}_0 + \hat{V})\,|\psi_n\rangle = E_n\,|\psi_n\rangle$, $\hat{H}_0\,|n\rangle = \varepsilon_n\,|n\rangle$
$E_n \simeq \varepsilon_n + \varepsilon_n^{(1)} + \varepsilon_n^{(2)} + \cdots$
$\varepsilon_n^{(1)} = \langle n|\hat{V}\,|n\rangle,\ \varepsilon_n^{(2)} = \sum_{m\neq n} \frac{|\langle m|\hat{V}\,|n\rangle|^2}{\varepsilon_n - \varepsilon_m}$

$$|\psi_n\rangle \simeq |n\rangle + \sum_{m\neq n} \frac{\langle m|\hat{V}\,|n\rangle}{\varepsilon_n - \varepsilon_m}\,|m\rangle$$

Time-dependent perturbation theory.
Transition probability $|i\rangle \to |f\rangle$
$P_{i\to f} = |d_{fi}(t)|^2$,
$d_{fi}(t) = \delta_{fi} - \frac{i}{\hbar}\int_{t_0}^{t} \mathrm{d}t'\langle f|\hat{V}(t')\,|i\rangle \mathrm{e}^{it'\omega_{fi}}$,
$\omega_{fi} = \frac{E_f - E_i}{\hbar}$.

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

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